

# Bayblend® FR3040 W

FR grades / Non reinforced

PC+ABS-FR(40)-Blend; flame retardant; for notebooks and thinwall applications

ISO Shortname

Pc+ABS-FR(40)

| Property                                            | Test Condition                        | Unit                    | Standard                     | typical Value |
|-----------------------------------------------------|---------------------------------------|-------------------------|------------------------------|---------------|
| <b>Rheological properties</b>                       |                                       |                         |                              |               |
| C Melt volume-flow rate                             | 240 °C/ 5 kg                          | cm <sup>3</sup> /10 min | ISO 1133                     | 23            |
| Melt viscosity                                      | 1000 s <sup>-1</sup> / 260 °C         | Pa·s                    | b.o. ISO 11443-A             | 250           |
| C Molding shrinkage, parallel                       | 60x60x2 mm/ 240 °C/ MT 80 °C/ 500 bar | %                       | ISO 294-4                    | 0.52          |
| C Molding shrinkage, normal                         | 60x60x2 mm/ 240 °C/ MT 80 °C/ 500 bar | %                       | ISO 294-4                    | 0.52          |
| <b>Mechanical properties (23 °C/50 % r. h.)</b>     |                                       |                         |                              |               |
| C Tensile modulus                                   | 1 mm/min                              | MPa                     | ISO 527-1,-2                 | 2650          |
| C Yield stress                                      | 50 mm/min                             | MPa                     | ISO 527-1,-2                 | 60            |
| C Yield strain                                      | 50 mm/min                             | %                       | ISO 527-1,-2                 | 4             |
| Stress at break                                     | 50 mm/min                             | MPa                     | ISO 527-1,-2                 | 57            |
| Strain at break                                     | 50 mm/min                             | %                       | b.o. ISO 527-1,-2            | 108           |
| Flexural modulus                                    | 2 mm/min                              | MPa                     | ISO 178                      | 2670          |
| Flexural strain at flexural strength                | 2 mm/min                              | %                       | ISO 178                      | 5.5           |
| Flexural stress at 3.5 % strain                     | 2 mm/min                              | MPa                     | ISO 178                      | 83            |
| Flexural strength                                   | 2 mm/min                              | MPa                     | ISO 178                      | 94            |
| Izod notched impact strength                        | 23 °C                                 | kJ/m <sup>2</sup>       | ISO 180/A                    | 48            |
| Izod notched impact strength                        | -30 °C                                | kJ/m <sup>2</sup>       | ISO 180/A                    | 12            |
| Izod notched impact strength                        | 23 °C                                 | kJ/m <sup>2</sup>       | ISO 21305/based on ISO 180/A | 71            |
| Izod notched impact strength                        | 0 °C                                  | kJ/m <sup>2</sup>       | ISO 21305/based on ISO 180/A | 68            |
| C Charpy impact strength                            | 23 °C                                 | kJ/m <sup>2</sup>       | ISO 179/1eU                  | N             |
| C Charpy notched impact strength                    | 23 °C                                 | kJ/m <sup>2</sup>       | ISO 179/1eA                  | 50            |
| C Charpy notched impact strength                    | -30 °C                                | kJ/m <sup>2</sup>       | ISO 179/1eA                  | 9             |
| C Puncture energy                                   | 23 °C                                 | J                       | ISO 6603-2                   | 49            |
| Ball indentation hardness                           | New                                   | N/mm <sup>2</sup>       | ISO 2039-1                   | 126           |
| <b>Thermal properties</b>                           |                                       |                         |                              |               |
| C Temperature of deflection under load              | 1.80 MPa                              | °C                      | ISO 75-1,-2                  | 87            |
| C Temperature of deflection under load              | 0.45 MPa                              | °C                      | ISO 75-1,-2                  | 96            |
| C Vicat softening temperature                       | 50 N; 50 °C/h                         | °C                      | ISO 306                      | 101           |
| Vicat softening temperature                         | 50 N; 120 °C/h                        | °C                      | ISO 306                      | 103           |
| C Coefficient of linear thermal expansion, parallel | 23 to 55 °C                           | 10 <sup>-4</sup> /K     | ISO 11359-1,-2               | 0.57          |
| C Coefficient of linear thermal expansion, normal   | 23 to 55 °C                           | 10 <sup>-4</sup> /K     | ISO 11359-1,-2               | 0.56          |
| C Burning behavior UL 94 [UL recognition]           | 0.8 mm                                | Class                   | UL 94                        | V-0           |
| C Burning behavior UL 94 [UL recognition]           | 0.75 mm                               | Class                   | UL 94                        | V-0 (bk)      |
| C Burning behavior UL 94-5V [UL recognition]        | 1.5 mm                                | Class                   | UL 94                        | 5VB           |
| Glow wire test (GWFI) [UL recognition]              | 1.0 mm                                | °C                      | IEC 60695-2-12               | 960           |
| <b>Electrical properties (23 °C/50 % r. h.)</b>     |                                       |                         |                              |               |
| C Volume resistivity                                |                                       | Ohm·m                   | IEC 60093                    | 1E15          |
| C Surface resistivity                               |                                       | Ohm                     | IEC 60093                    | 1E17          |
| <b>Other properties (23 °C)</b>                     |                                       |                         |                              |               |
| C Water absorption (saturation value)               | Water at 23 °C                        | %                       | ISO 62                       | 0.14          |
| C Density                                           |                                       | kg/m <sup>3</sup>       | ISO 1183-1                   | 1190          |
| <b>Processing conditions for test specimens</b>     |                                       |                         |                              |               |
| C Injection molding - Melt temperature              |                                       | °C                      | ISO 294                      | 240           |
| C Injection molding - Mold temperature              |                                       | °C                      | ISO 294                      | 80            |

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|-----------------------------------------------------|----------------|------|----------|---------------|
| <b>Recommended processing and drying conditions</b> |                |      |          |               |
| Melt temperatures                                   |                | °C   | -        | 250-280       |
| Standard Melt temperature                           |                | °C   | -        | 260           |
| Barrel Temperatures - Rear                          |                | °C   | -        | 240-260       |
| Barrel Temperatures - Middle                        |                | °C   | -        | 250-270       |
| Barrel Temperatures - Front                         |                | °C   | -        | 260-280       |
| Barrel Temperatures - Nozzle                        |                | °C   | -        | 260-280       |
| Mold Temperatures                                   |                | °C   | -        | 60-80         |
| Hold Pressure (% of injection pressure)             |                | %    | -        | 50-75         |
| Plastic Back Pressure (specific)                    |                | bar  | -        | 50-100        |
| Peripheral Screw Speed                              |                | m/s  | -        | 0.1-0.3       |
| Shot-to-Cylinder Size                               |                | %    | -        | 30-70         |
| Dry Air Drying Temperature                          |                | °C   | -        | 85            |
| Dry Air Drying Time                                 |                | h    | -        | 4             |
| Moisture Content max. (%)                           |                | %    | -        | 0.02          |
| Vent Depth                                          |                | mm   | -        | 0.02-0.04     |

**C** These property characteristics are taken from the CAMPUS plastics data bank and are based on the international catalogue of basic data for plastics according to ISO 10350.

Impact properties: N = non-break, P = partial break, C = complete break

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## Disclaimer

### Typical value

These values are typical values only. Unless explicitly agreed in written form, they do not constitute a binding material specification or warranted values. Values may be affected by the design of the mold/die, the processing conditions and coloring/pigmentation of the product. Unless specified to the contrary, the property values given have been established on standardized test specimens at room temperature.

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### Recommended Processing and Drying Conditions

Barrel temperatures are valid for a standard 3-zone barrel. Temperature set-up for different barrel types may change according to configuration. Values for hold pressure as percentage of injection pressure may vary depending on, amongst others, part geometry, injection molding machine and injection mold. Drying conditions are for dry air dryers only. Drying times and drying temperatures may differ depending on valid dryer type. Further information is provided by your local Covestro support as well as in the following brochures: Injection Molding of High Quality Molded Parts - Drying; Determining the Dryness of Makrolon by TVI Test; The fundamentals of shrinkage in thermoplastics; Shrinkage and deformation of glass fiber reinforced thermoplastics [...]. <https://www.plastics.covestro.com/Library/Overview.aspx>

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Covestro AG  
Polycarbonates Business Unit  
Kaiser-Wilhelm-Allee 60  
51373 Leverkusen  
Germany  
[plastics@covestro.com](mailto:plastics@covestro.com)  
[www.plastics.covestro.com](http://www.plastics.covestro.com)